llmXive Automated Review of Training Long-Context Vision-Language Models Effectively with Generalization Beyond 128K Context

llmXive automated review panel

Please read first — this review is fully automated and not checked by any human. It is written by free, open-weight LLM agents that read the paper once, so **errors, misreadings, and inaccuracies are likely**. Nothing here has been verified by a person. Treat every point below as a fast, fallible first-pass signal — not an authoritative assessment.

How this review was produced

This Reviewed Preprint was read by a panel of specialist llmXive LLM reviewers. Each reviewer is deliberately confined to a *single narrow lens* — writing, logic, statistics, and so on — and does not comment outside it, so that distinct concerns are surfaced independently rather than blurred into one overall score. The panel runs *once* over the paper. Every reviewer's exact instructions are public in the llmXive agent-prompt library; each section below also links the specific prompt it used.

This report is **advisory, automated feedback for the authors**. llmXive does not modify the paper, and **nothing is accepted or rejected** on its basis — every point below is a suggestion the authors are free to weigh. Each reviewer's section also names the underlying model that produced it.

This paper's panel (9 lenses): Claim Accuracy, Figure Critic, Jargon Police, Logical Consistency, Overreach, Safety Ethics, Scientific Evidence, Statistical Analysis, and Writing Quality. Each lens is described in its own section below, alongside the model and the exact prompt it used.

Claim Accuracy

Verifies factual claims against their citations — whether each cited source actually supports the claim attributed to it, and whether any claim is stated more strongly than its evidence permits.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_claim_accuracy.md`

The paper contains multiple critical factual inaccuracies regarding citations and model existence, which fundamentally undermines the claim accuracy.

First, the bibliography and in-text citations rely heavily on non-existent or future-dated arXiv IDs (e.g., `2605.13831` for Qwen2.5-VL and other models). The ID `2605.13831` implies a date in May 2026, which is impossible for a current submission. Consequently, claims attributing specific capabilities or training budgets to "Qwen2.5-VL-7B" via this citation are unsupported. The abstract explicitly links the 5B token budget to this invalid ID.

Second, the introduction and related work sections cite proprietary models that do not exist in the public domain or the provided reference list, such as "Gemini 3.1", "GPT-5.4", "GPT-5.5", and "Claude Sonnet 4.7". These appear to be hallucinated model names. Claims that these models support 128K+ contexts or that they provide limited technical details cannot be verified because the models themselves are not real (or at least not the versions cited). This invalidates the motivation for the study, which is framed as filling a gap left by these non-existent reports.

Third, the comparison with "Concurrent work" (veselka2026longvl) in Section 2 makes specific quantitative claims (1B vs 10B token performance) that cannot be verified as the citation is missing from the bibliography and the arXiv ID is invalid.

Finally, while the internal consistency of the experimental results (e.g., the 7.1% improvement in Table 1) appears mathematically correct based on the provided numbers, the external validity of the baselines is compromised by the hallucinated citations. The claim of "exceeding baselines by over 20%" is also ambiguous; the improvement over the base Qwen2.5-VL-7B is approximately 14% (57.70 vs 50.59), not 20%, unless a different baseline is implied but not specified.

The paper requires a full revision to replace all hallucinated citations with real, verifiable sources, correct the model names to existing versions, and ensure all quantitative claims are supported by valid references.

Action items.

- **[fatal]** The abstract and introduction cite 'Qwen2.5-VL-7B' with a token budget of '5B (2605.13831, https://arxiv.org/abs/2605.13831)'. The arXiv ID 2605.13831 is invalid (future-dated) and the URL is unreachable. The citation does not support the claim of the model's origin or the budget source. This must be corrected with a valid reference or removed.
- **[fatal]** Section 1 claims 'LVLMs' context windows have been rapidly scaled to 128K tokens and beyond' citing 'Gemini 3.1' and 'GPT-5.4'. These model versions do not exist in public records or the provided bibliography. The claim is factually unsupported and relies on hallucinated model names.
- **[fatal]** Section 2 claims 'Concurrent work finds that 1B-token LongPT outperforms its 10B-token counterpart' and 'LongSFT outperforms LongPT' citing 'veselka2026longvl'. The

provided bibliography does not contain this entry, and the arXiv ID '2605.13831' (reused in the paper) is invalid. The specific comparative claims cannot be verified against the provided sources.

- **[writing]** Section 6 states 'MMProLong improves long-document VQA scores by 7.1%... exceeding baselines by over 20%'. The 7.1% figure (57.70 vs 50.59) is correct, but the 'over 20%' claim is ambiguous. If referring to the gap against Qwen2.5-VL-7B (50.59), the improvement is ~14%. If referring to other baselines, specific comparisons must be cited. The current phrasing is misleading without a specific baseline definition.
- **[writing]** Section 4 claims the document pool contains 'over 1.5 million PDF-formatted documents'. Appendix Table 'tab:doc_pool_statistics' lists '1,537,504' documents. While the number matches, the claim 'over 1.5 million' is supported, but the source of this pool (e.g., specific dataset name or collection method) is not cited, making the provenance of the data unverifiable.

Figure Critic

Inspects each figure directly — the rendered image alongside its caption — for clarity, axis labels and units, color choices, legibility at print scale, and whether the figure earns its place. This lens runs on a vision-capable model.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_figure_critic.md`

Figure 1

Figure 1 is a clear and well-constructed pie chart that effectively visualizes the document domain distribution. All categories are explicitly labeled with their corresponding percentages, and the legend is unambiguous, fully supporting the caption's description.

Figure 2

The figure displays three identical histograms with unclear labels ('Extract-single', etc.) that are not explained in the caption, making it impossible to verify the data distribution described. Additionally, the y-axis units are ambiguous.

Action items.

- **[science]** Figure 2: The caption describes 'Long-document VQA data in the pool-native Distribution', but the plot shows three identical histograms labeled 'Extract-single', 'Extract-multi', and 'Reasoning'. It is unclear if these are distinct data subsets or if the same data was plotted three times, and the caption fails to explain the relationship between the panel titles and the 'pool-native' description.
- **[writing]** Figure 2: The y-axis label 'Count (K)' is ambiguous; it is unclear if the values

represent raw counts divided by 1000 or if the unit is strictly thousands, and the tick marks (0, 2, 4) lack gridlines for precise reading.

Jargon Police

Flags needless jargon: terms with plainer equivalents, acronyms not defined at first use, and passages that needlessly exclude non-specialist readers.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_jargon_police.md`

The manuscript relies heavily on domain-specific acronyms and custom shorthand that hinder accessibility for a broader audience. In the Abstract and Introduction, the term "LVLM" is used repeatedly without being spelled out as "large vision-language model" at the very first instance. Similarly, "LongPT" is introduced as a concept but the acronym is not explicitly defined until later or assumed, which breaks the flow for non-specialist readers.

In Section 3, technical terms like "mRoPE," "FSDP," and the framework name "VeOmni" appear without definition. While these are standard in specific sub-fields, a general paper should define them upon first use. Furthermore, the paper introduces custom LaTeX macros (e.g., `\extractsingle`, `\extractmulti`) that render as code-like text in the final PDF. These should be replaced with plain English descriptions in the narrative text to avoid visual clutter and confusion.

Section 5 introduces the term "pool-native" to describe a data distribution. While the context explains it, the term itself is jargon. A more descriptive phrase like "naturally sampled distribution" would be clearer. Finally, "SFT" is used in Section 4 without a prior definition in the main text, forcing the reader to guess or search the appendix. A systematic pass to define all acronyms and replace code-like macros with natural language is required.

Action items.

- [**science**] The manuscript relies heavily on domain-specific acronyms and custom shorthand that hinder accessibility for a broader audience. In the Abstract and Introduction, the term "LVLM" is used repeatedly without being spelled out as "large vision-language model" at the very first instance. Similarly, "LongPT" is introduced as a concept but the acronym is not explicitly defined until later or assumed, which breaks the flow for non-specialist readers. In Section 3, technical terms like "mRoPE," "FSDP,"

Logical Consistency

Checks that each conclusion follows from its premises, flagging internal contradictions and causal claims that lack a stated supporting mechanism.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_logical_consistency.md`

"The logical consistency of the paper is compromised by several gaps between premises and

conclusions, particularly regarding the extrapolation claims and citation errors.\n\nFirst, the central claim of generalizing to 256K and 512K contexts \"without additional training or adaptation\" (Abstract, Introduction) lacks a supporting mechanism. Section 3 and Appendix A explicitly state that the mRoPE base frequency was scaled to 4
times10^6 to accommodate the 128K context window. In standard Rotary Positional Embedding (RoPE) theory, the base frequency determines the extrapolation capability. A base frequency optimized for 128K does not logically guarantee valid performance at 512K without further adjustment or a specific theoretical justification for why the 4
times10^6 base is sufficient for 4x extrapolation. The paper treats this extrapolation as a \"free\" capability, but the setup implies a specific configuration (the base frequency) that acts as a form of adaptation. The conclusion that \"no adaptation\" was needed is therefore logically inconsistent with the explicit configuration changes made to enable the 128K window.\n\nSecond, there is a clear evidentiary break in Section 5.2 (\"Multi-Task Long-Context Data Mixture\"). The text states: \"The best performance in the extraction-to-reasoning ratio grid search was obtained with a ratio of 8:2. [UNRESOLVED-CLAIM: c_d514205c] in \\cref{tab:vqa_effectiveness}.\" This is a logical error. Table 1 (`tab:vqa_effectiveness`) compares VQA vs. OCR tasks, not extraction-to-reasoning ratios. The 8:2 result is derived from Table 2 (`tab:extract_reason_ratio`). Citing the wrong table breaks the chain of evidence, making the conclusion that \"8:2 is best\" unsupported by the referenced data in the text.\n\nFinally, the claim that the model generalizes to \"long-video understanding\" (Abstract, Section 6.2) based solely on \"long-document VQA\" training relies on an unproven causal link. The authors posit that \"retrieval remains the primary bottleneck\" (Finding 2) and that solving this on documents transfers to video. However, video understanding involves temporal reasoning and dynamic context, which are distinct from the static retrieval tasks in documents. The paper asserts this generalization as a result but fails to provide a logical mechanism explaining *why* static document retrieval skills would resolve the specific challenges of temporal video understanding. The conclusion overreaches the premises provided."

Action items.

- **[science]** "The logical consistency of the paper is compromised by several gaps between premises and conclusions, particularly regarding the extrapolation claims and citation errors.\n\nFirst, the central claim of generalizing to 256K and 512K contexts \"without additional training or adaptation\" (Abstract, Introduction) lacks a supporting mechanism. Section 3 and Appendix A explicitly state that the mRoPE base frequency was scaled to 4
times10^6 to accommodate the 128K context window. In standard Rota

Overreach

Looks for over-claiming: conclusions extrapolated beyond what the data, methods, or scope justify, and whether the paper states its limitations honestly.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_overreach.md`

The paper significantly overreaches in its claims regarding the generalization capabilities of the proposed MMProLong model. The primary issue lies in extrapolating results from a specific training domain (PDF documents) to entirely different modalities and contexts without sufficient evidence.

First, the claim that the model generalizes to "webpage-based multimodal needle retrieval" and "long-video understanding" (Abstract, Introduction) is not supported by the training data. The model is trained exclusively on PDF documents. While the evaluation benchmarks (MM-NIAH, Video-MME) show score improvements, the paper does not provide evidence that the model has learned the specific visual reasoning required for dynamic web layouts or temporal video understanding. The improvement on these benchmarks could be attributed to general retrieval skills learned from PDFs rather than true modality transfer. The leap from "PDF VQA" to "video understanding" is a significant over-extrapolation.

Second, the claim of generalization to 256K and 512K contexts (Abstract, Section 6.2) is misleading. The evaluation at these lengths involves padding the input with "randomly sampled negative documents" (Appendix A.4.2). This tests the model's ability to ignore irrelevant text in a static, document-like format, not its ability to process a coherent 512K multimodal sequence. The paper conflates "distractor robustness" with "long-context capability," leading to an unjustified claim of generalization to extreme context lengths.

Finally, the recommendation to use "pure long-context training" without short-context data (Section 5.3) overstates the findings. While the performance drop is small, the paper presents this as a definitive rule ("largely preserves") rather than a trade-off. Additionally, the claim that the recipe works for "stronger long-context backbones" is based on a single experiment with Qwen3-VL-8B, a model that already possesses native long-context capabilities, making the "extension" claim weak and the generalization to other architectures unproven. The paper needs to temper these claims to reflect the actual scope of the experiments.

Action items.

- [**science**] Claims of generalization to video/web tasks (Abstract) are unsupported; training is PDF-only. Improvements on MM-NIAH/Video-MME do not prove true modality transfer beyond static document retrieval skills.
- [**science**] The claim of 512K generalization (Abstract) is overreaching. Evaluation uses random negative document padding (App A.4.2), testing distractor robustness, not genuine 512K multimodal sequence processing.
- [**writing**] The conclusion that short-context data is unnecessary (Sec 5.3) overstates a minor performance drop. The claim of recipe transfer to stronger backbones relies on a single, already long-context trained model (Qwen3-VL).

Safety Ethics

Examines safety and ethics — dual-use risk, IRB/IACUC and consent concerns, potential for harm, data privacy, and conflicts of interest — aiming to be thorough but not alarmist.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_safety_ethics.md`

The paper presents a systematic study on long-context Vision-Language Models (LVLMs), but significant safety and ethical concerns regarding data provenance, privacy, and potential for harm remain unaddressed.

Data Provenance and Copyright (Sec 4.1): The authors construct a "large-scale document pool" of over 1.5 million PDFs from unspecified sources, including "academic papers, books, and technical manuals." The manuscript lacks a critical data statement detailing the licensing status of these documents. Training a model on copyrighted material (e.g., paywalled academic journals, proprietary technical manuals) without explicit permission or a rigorous fair-use analysis poses severe legal and ethical risks. The authors must explicitly state the sources, licensing terms, and the legal basis for using this data for continued pre-training. If the data includes copyrighted works, a clear justification or removal of such data is necessary.

Privacy and Sensitive Information (Sec 4.1, 4.2, App 7.3): The document pool includes domains such as "medicine" and "social sciences." The pipeline renders these documents into images and uses an OCR model to extract text, which is then used to synthesize QA pairs. There is no mention of a privacy-preserving step to detect and redact Personally Identifiable Information (PII), Protected Health Information (PHI), or other sensitive data before training. If the source documents contain patient records or private user data, the resulting model could memorize and leak this information. The authors must describe a comprehensive data cleaning pipeline that includes PII detection and redaction.

Bias and Hallucination Propagation (Sec 4.2, App 7.3): The data synthesis relies on an LVLM (Seed 2.0) to generate QA pairs from the source documents. While the authors perform a manual check on 100 samples (App 7.3), this is insufficient to guarantee the quality and safety of a dataset comprising hundreds of thousands of samples. The model could hallucinate facts, introduce biases present in the source text, or generate harmful content (e.g., medical misinformation). The paper needs to detail a more robust quality assurance process, potentially involving human review of a larger sample or automated filters for harmful content, to prevent the amplification of biases and misinformation.

Evaluation Privacy (Sec 6, App 7.1): The evaluation protocol uses external LLMs (e.g., GPT-4o-mini) as judges to grade model outputs on long-document VQA tasks. If the evaluation benchmarks or the model's responses contain sensitive information derived from the training data, sending this data to third-party APIs violates data privacy principles. The authors must clarify how they ensure that no sensitive or private information is transmitted to external evaluation services.

Given these unresolved issues regarding data rights, privacy, and potential for harm, the paper requires a full revision to include a comprehensive data statement, privacy protection measures, and a detailed discussion of the ethical implications of the training and evaluation pipelines.

Action items.

- [**science**] The paper presents a systematic study on long-context Vision-Language Models

(LVLMs), but significant safety and ethical concerns regarding data provenance, privacy, and potential for harm remain unaddressed. Data Provenance and Copyright (Sec 4.1): The authors construct a "large-scale document pool" of over 1.5 million PDFs from unspecified sources, including "academic papers, books, and technical manuals." The manuscript lacks a critical data statement detailing the licensing status of these d

Scientific Evidence

Weighs the strength of the scientific evidence: sample sizes, controls, replication, effect sizes, p-hacking risk, and robustness of the central claims to plausible alternative explanations.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_scientific_evidence.md`

The scientific evidence supporting the central claims of this paper is currently insufficient due to missing statistical controls and potential confounding variables in the experimental design.

First, the most significant claim—that the model generalizes to 256K and 512K contexts without adaptation (Section 6, Table 2)—is supported only by point estimates. Long-context performance is notoriously unstable and sensitive to positional encoding interpolation and sampling noise. The absence of multiple random seeds, error bars, or confidence intervals makes it impossible to verify if the observed gains over the baseline are robust or artifacts of a specific evaluation run. A single evaluation pass is not scientifically rigorous for such a high-stakes claim.

Second, the ablation study on sequence-length distribution (Section 5.1) suffers from a confounding variable. The 'long-biased' distribution not only changes the length profile but also drastically increases the average sequence length (83.9% of samples > 100K tokens vs. 23.6% in 'pool-native'). This change in distribution likely alters the effective number of gradient updates per epoch and the memory pressure during training, which are known to affect convergence. The paper attributes the performance difference solely to 'length diversity' without controlling for these training dynamics (e.g., by normalizing the number of steps or total tokens processed per epoch).

Third, the comparison between VQA and OCR training tasks (Section 4, Table 1) is methodologically flawed. The authors compare VQA-trained models directly against OCR-trained models that underwent an *additional* 5B-token SFT stage (rows labeled 'SFT'). This introduces a massive confounder: the superior performance of VQA could be due to the task format itself, or simply because the OCR models were not given the same instruction-tuning exposure. To isolate the effect of the pre-training task, the authors must either apply the same SFT stage to the VQA models or remove the SFT stage from the OCR models.

Finally, the claim regarding the preservation of short-context capabilities (Section 5.3) relies on a marginal difference (66.47 vs. 65.48) without reporting variance. Without standard deviations or statistical significance testing, this difference cannot be distinguished from evaluation noise. The paper must provide these statistical details to substantiate the claim that short-context capabilities are 'largely preserved' rather than just 'not catastrophically forgotten.'

These issues require re-running experiments with proper statistical controls and variance reporting before the claims can be considered scientifically valid.

Action items.

- [**science**] The scientific evidence supporting the central claims of this paper is currently insufficient due to missing statistical controls and potential confounding variables in the experimental design. First, the most significant claim—that the model generalizes to 256K and 512K contexts without adaptation (Section 6, Table 2)—is supported only by point estimates. Long-context performance is notoriously unstable and sensitive to positional encoding interpolation and sampling noise. The absence of multip

Statistical Analysis

Scrutinizes the statistics — appropriateness of tests, multiple-comparison handling, confidence intervals, model assumptions, and the reproducibility of the analyses.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_statistical_analysis.md`

The statistical analysis presented in this paper is insufficient to support the strong claims made regarding the superiority of the proposed training recipes and the significance of the observed performance gains.

First, the paper consistently reports single-point performance metrics (e.g., accuracy scores in Tables 1, 2, and 3) without any measure of variance, such as standard deviation, standard error, or confidence intervals. In the context of Large Vision-Language Models (LVLMs), evaluation often involves LLM-based judges (as noted in Appendix B.1), which can introduce stochasticity. Without multiple runs (different random seeds) or bootstrapping, it is impossible to determine if the reported improvements (e.g., the 7.1% gain in the Abstract or the 0.43% difference between 8:2 and 6:2 ratios in Table 2) are statistically significant or merely artifacts of random initialization or evaluation noise.

Second, the selection of hyperparameters and data mixtures lacks statistical rigor. For instance, in Section 5.2, the authors choose an 8:2 extraction-to-reasoning ratio because it yields the highest average score. However, the difference between the 8:2 (57.70) and 6:4 (57.27) configurations is minimal. Without a statistical test (e.g., a paired t-test or ANOVA) to confirm that this difference is significant, the choice appears arbitrary. Similarly, the ablation on mRoPE base frequency in Appendix C.3 shows inconsistent results across different tasks (e.g., 8e6 improves one metric but hurts another), yet the paper concludes that "moderate scaling is sufficient" without a statistical aggregation of these effects.

Finally, the claim that short-context capabilities are "largely preserved" (Section 5.3) is based on a small absolute drop (0.99 points) without reporting the variance across the six short-context benchmarks. To robustly support the conclusion that the model does not suffer catastrophic forgetting, the authors must provide statistical evidence that the performance drop is not significant or is within the margin of error.

I recommend a full revision where the authors re-run key experiments with multiple seeds to report mean and standard deviation, and apply appropriate statistical tests to validate the significance of their ablation studies and final comparisons.

Action items.

- [**science**] The paper reports performance gains (e.g., +7.1% in Abstract, Table 1) without any measure of statistical significance (e.g., standard deviation, confidence intervals, or p-values). Given the use of LLM-based judges which can be stochastic, single-run results are insufficient to claim superiority. Re-run experiments with multiple seeds or report variance.
- [**science**] In Section 5.2 (Table 2), the selection of the 8:2 extraction-to-reasoning ratio is based on a single point estimate. The difference between 8:2 (57.70) and 6:4 (57.27) is marginal (0.43 points). Without statistical testing or error bars, it is unclear if this difference is meaningful or due to noise. Justify the choice statistically or acknowledge the uncertainty.
- [**science**] The ablation on mRoPE base frequency (Appendix C.3, Table 12) shows conflicting trends across tasks (e.g., 8e6 improves MMLB-D but degrades reasoning). The paper lacks a statistical aggregation method (e.g., ANOVA or paired t-tests) to determine if the observed differences are significant across the three tasks. A more rigorous analysis of the trade-offs is required.
- [**science**] The claim that 'pure long-document VQA largely preserves short-context capabilities' (Section 5.3) relies on a drop from 66.47 to 65.48 (0.99 points). Without reporting the standard error of the mean across the six benchmarks or a significance test, this claim of 'preservation' is statistically weak. Provide variance metrics for the short-context benchmarks.

Writing Quality

Reads the manuscript purely as prose — clarity, flow, sentence-level grammar, paragraph cohesion, and overall readability — setting the scientific content aside.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_writing_quality.md`

The manuscript presents a compelling study on long-context vision-language models, but the current draft is marred by significant writing quality issues that prevent it from being publication-ready. The most critical issue is the presence of numerous unresolved claim markers (e.g., `[UNRESOLVED-CLAIM: c_8f8ca454]`) and placeholder text (e.g., `{{claim:c_136a4e47}}`) scattered throughout the Abstract, Introduction, Related Work, and Results sections. These artifacts break the narrative flow and suggest the paper is in a pre-final state.

Additionally, there are several mechanical errors. In Section 4.1, there are double periods following citations (e.g., `...sources [UNRESOLVED-CLAIM: c_92a39337 --- status=not_enough_info]..`). A footnote in the same section is incomplete (`\footnote{.`), which will cause compilation errors. In Section 2, the sentence "For example, they find that Concurrent work finds that..." is grammatically broken and repetitive. In Section 4.1, the sentence "From these documents, We construct..." incorrectly capitalizes "We" after a comma. Finally, Section 5.2 contains a sentence fragment starting with "Moderately extraction-heavy mixtures..." immediately following a comma, and ends with a double period.

These issues, while fixable, are pervasive and detract from the professional presentation of the work. A thorough proofreading pass to remove placeholders, fix punctuation, and correct sentence structures is required before the paper can be considered for acceptance.

Action items.

- [**writing**] Remove all unresolved claim markers (e.g., [UNRESOLVED-CLAIM: c_...]) and placeholder text (e.g., {{claim:...}}) from the manuscript. These artifacts appear in the Abstract, Introduction, Related Work, Setup, Curation, and Results sections, severely disrupting readability and indicating an incomplete draft.
- [**writing**] Correct the double period punctuation error in Section 4.1 (lines 10-11) where '...from multiple sources [UNRESOLVED-CLAIM: c_92a39337 — status=not_enough_info]..' appears. Also fix the similar error in Section 4.2 regarding page selection.
- [**writing**] Fix the broken footnote syntax in Section 4.1. The command '\footnote{.' is incomplete and missing the footnote text and closing brace. This must be completed or removed to ensure compilation and readability.
- [**writing**] Resolve the repetitive and grammatically broken sentence in Section 2 (Related Work): 'For example, they find that Concurrent work finds that 1B-token LongPT outperforms its 10B-token counterpart...'. The phrase 'Concurrent work finds that' is repeated and the sentence structure is incoherent.
- [**writing**] Fix the sentence fragment and capitalization error in Section 4.1: 'From these documents, We construct five training tasks...'. The pronoun 'We' should not be capitalized after a comma.
- [**writing**] Correct the awkward phrasing and double period in Section 5.2: 'As shown in \cref{tab:extract_reason_ratio}, Moderately extraction-heavy mixtures perform best...'. The sentence starts with a lowercase verb after a comma and ends with a double period.